

Emotion Recognition from Biometric Signals

Lorenzo Bualetti

s353632@studenti.polito.it

Tommaso Franzoni

s359190@studenti.polito.it

Alex Martinelli

s357881@studenti.polito.it

Abstract

This project explores the development and evaluation of deep learning architectures for the binary classification of emotional states (positive vs. negative valence) utilizing psychophysiological signals. Leveraging the POPANE dataset, which provides high-resolution recordings of ECG, EDA, and affective features from 1,157 subjects, the study investigates the transition from unimodal to multimodal systems to overcome the inherent limitations of single-signal analysis.

The research is structured into two primary experimental phases. In the first phase, a 1D-CNN architecture was optimized for ECG signal processing. Initial iterations revealed a significant statistical bias toward the majority class due to dataset imbalance. Through the implementation of weighted loss functions, enhanced regularization (Dropout 0.4, L2 decay), and final physical undersampling, an unbiased unimodal baseline was established with a Macro F1-Score of 0.63. SHAP interpretability analysis confirmed that while the model correctly identifies QRS complex morphology, it fails to disambiguate high-arousal emotions of opposite valence.

The second phase introduced multimodal integration through Early and Late Fusion strategies. Experimental results demonstrate that Early Fusion, which integrates features at the input level to capture cross-modal temporal correlations, is the superior strategy, achieving an optimal Macro F1-Score of 0.6630. Conversely, Late Fusion exhibited a higher risk of "predictive collapse" and "lazy learning" shortcuts, requiring independent branch optimization and majority voting to reach stability.

The final results validate that while cardiac dynamics contain foundational emotional information, the integration of electrodermal activity is essential to resolve arousal-related ambiguities. This project provides a robust framework for Affective Computing, highlighting the critical role of data rebalancing and inter-modal feature extraction in developing generalized emotional representation models for intelligent cyber-physical systems.

1 Introduction

It is now well established that biometric signals from the heart are influenced by an individual's specific emotions. Several psychological and neuroscientific studies demonstrate a strong correlation between emotional responses and objective physiological metrics, such as cardiovascular, electrodermal, and respiratory measures.

First of all, emotions can be divided in two primary categories based on valence: positive (e.g. joy, amusement, or serenity) and negative (e.g. fear, anger, or sadness). Subsequently, they can be classified by their level of intensity: high-arousal emotions, which are associated with greater electrical activity, and low-arousal emotions, which do not exhibit the same level of intensity. Within this framework, it is crucial to note that opposite valences can share the same physiological signal pattern: an intense negative experience can elicit similar neurological measurements to joy or enthusiasm.

Recently, emotions have gained significant attention in fields such as Computer Science and Engineering. An example is given by Affective Computing, a branch of AI technology that utilizes physiological and neurological measurements to develop algorithms capable of detecting and profiling the user experience. Machine learning models can identify patterns of biometric signals to understand the emotional state of an individual. This capability represents one of the most intriguing aspects of modern AI, proving highly valuable across various fields, including healthcare, psychology, and education.

Various types of biometric signals can be acquired from patients using different sensors and detection systems. The model employed can use a single signal to classify the patient's emotions. While this unimodal approach is the fastest and simplest to implement, it is generally the least accurate in predicting emotional states. Conversely, a multimodal approach, using multiple physiological measurements, yields more robust estimations but may incur higher latency. Furthermore, some metrics may not be useful for the model, which results in redundant preprocessing and a waste of computational power and resources.

The main objective of this project is to develop a model capable of detecting and classifying an individual's emotional state. To achieve this, various features extracted from biometric signals were utilized. The study is structured into two main phases: one using a unimodal system and the other using a multimodal system, incorporating three different metrics. Within the multimodal framework, two different data integration techniques were applied: Early Fusion and Late Fusion. Moreover, to address the "black-box" nature of these models and validate the biological relevance of the predictions, SHAP values were computed. Finally, we compare the three sets of

results to identify the best-performing trained model.

The remainder of this paper is organized as follows: Section 2 describes the dataset and the preprocessing stage; Section 3 details the architecture and the methods applied to develop the network; Section 4 illustrates the results obtained from the training phase; Section 5 concludes the work.

2 Dataset

The complete data source is the POPANE Dataset [1], which contains psychophysiological data from seven large-scale experiments investigating the functions of positive and negative emotions. It comprises seven different studies testing how emotions influence cardiovascular parameters in different scenarios. This public and open-source dataset reports eight biosignals from 1157 subjects, with a gender distribution of 45% female and 55% male, all aged between 18 and 38.

The data repository is organized into seven ZIP archives, corresponding to the individual studies. Each of these directories contains a set of CSV files, where each file contains the physiological measurements of a single individual. The total dimension of the dataset amounts to 150 GB.

Within each folder, the values of the biometric signals and metadata, corresponding to a specific emotional state of a single participant, are stored in separate files. For example, the file "25_gratitude.csv" contains the physiological recordings of participant 25 while experiencing gratitude. However, all data for a given individual is also aggregated into a comprehensive file named "<id>_all.csv". Consequently, much of the information in this dataset is redundant. We were therefore able to discard the specific emotion files, utilizing only the file that contains the complete physiological profile of each subject.

The consolidated files feature a 10-line metadata header containing personal details of the patient (e.g. age and gender) and technical specifications of the recording (e.g. sampling rate and device used). Since the majority of this metadata is irrelevant to the model, it was discarded during preprocessing. The final row of this block represents the actual data header, which indicates the specific biometric signals and provides a timestamp to ensure the strict temporal alignment. The final column, although not always present, is pivotal: it contains the categorical marker indicating the precise emotional state experienced at that timestamp. We utilized these marker-codes as the foundational ground truth to formulate a binary classification problem. Specifically, the diverse set of specific emotions recorded in the dataset was aggregated into two general classes: positive valence (1) and negative valence (0). By parsing the blocks of data, records associated with marker-codes reflecting positive affective states (e.g. joy, amusement, gratitude) were mapped to the positive class. Conversely,

segments corresponding to negative or stressful states (e.g. fear, stress, anger) were assigned to the negative class.

In order to reduce the amount of data loaded into RAM, handle the substantial volume of the dataset, and prevent memory overflow during the training phase, we implemented a lazy loading mechanism. This was achieved by creating an index file that enables random access to a window of samples in the dataset, ensuring that only the required segments are loaded into memory. This index file tracks the precise location of each physiological sequence and, crucially, segments the continuous biosignals into fixed-size temporal windows, each composed of 1000 units, transforming the raw continuous streams into discrete and fixed-length tensors, suitable for feature extraction in the network.

In a second step, the dataset was rigorously partitioned into training (70%), validation (15%), and testing subsets (15%). Once again, we utilized an index file to speed up and optimize data management. To prevent data leakage and guarantee that the model learns generalized emotional representations rather than patient-specific physiological signatures, a subject-wise splitting strategy was enforced. Consequently, all data windows belonging to a specific individual (based on their subject_id) were exclusively assigned to a single subset.

Regarding the selected biosignals, the unimodal architecture of the first phase utilized exclusively the Electrocardiogram (ECG). This signal represents the electrical activity of the heart over time and is measured in microvolts. For the multimodal experiments, Electrodermal Activity (EDA) and Affective features were incorporated into the model. EDA measures continuous variations in the electrical characteristics of the skin, specifically skin conductance driven by sympathetic nervous system arousal, and is quantified in microsiemens. Additionally, the affective features provide a mix of complementary behavioral metrics, expressed without a physical unit. Finally, as a last preprocessing step, we applied Z-normalization to standardize the scale of all biosignals.

During the model development phase, an analysis of the POPANE dataset revealed a significant class imbalance: the positive class contained 14,400 instances, whereas the negative class contained only 4,500. To address this discrepancy and prevent the neural network from developing a statistical bias towards the majority class, we decided, at a specific point of the training process, to rebalance the dataset. Although data augmentation is typically the optimal solution for this issue, our limited storage and computational resources forced us to use the opposite approach: undersampling. We randomly discarded approximately 66% of the samples belonging to the positive majority class. This preprocessing step resulted in a perfectly balanced final dataset, consisting of an equal number of positive and negative temporal windows.

3 Methods

3.1 Network Architecture, Model, and Loss Used

In the first phase, the CNN architecture deployed for ECG classification mirrors the structure shown in Figure 1.

The network is composed of two primary modules: a feature extraction block and a classification head. The feature extractor utilizes convolutional layers with a kernel size of 7 to isolate relevant biosignal patterns and mitigate noise. These are coupled with pooling layers that progressively reduce the spatial dimensionality of the feature maps while preserving salient information. Following extraction, the feature maps are flattened and passed into the fully connected layers of the classification block. Its purpose is to estimate the corresponding emotional state. During training, we chose not to modify the architecture in order to maintain low computational costs due

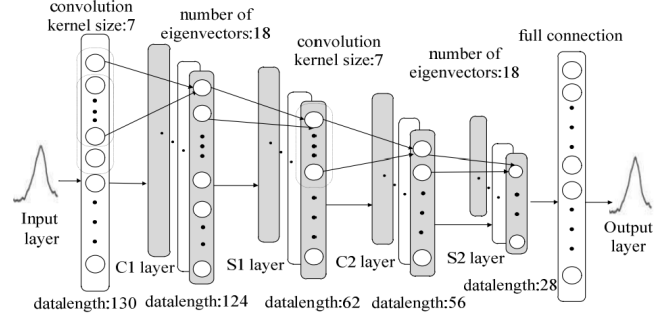


Figure 1: 1D CNN Architecture used [2]

to our hardware constraints. Our optimization efforts were instead exclusively focused on hyperparameter tuning.

Building upon this baseline, the second phase required the adaptation of the model to support multimodal inputs (ECG, EDA, and affective features). In the early fusion configuration, the normalized input signals were concatenated at the input level, enabling a single unified 1D-CNN to process them simultaneously and capture cross-modal correlations from the initial states in the network. Conversely, the late fusion architecture utilized independent 1D-CNN branches for each signal. The respective outputs were then fused in a final fully connected layer, performing a majority voting mechanism: for instance, if two branches predicted class 0 and one predicted class 1, the final model output would be 0.

Since the objective was framed as a binary classification problem, the final network variants were optimized utilizing a Binary Cross-Entropy (BCE) loss function. However, during preliminary experiments, before implementing the undersampling strategy to balance the class distribution, we used a Weighted BCE to address the dataset imbalance, discussed in the previous section. This approach incorporates a class-weighting mechanism that adjusts the loss penalty: yielding an incorrect prediction on an instance of the minority class incurs a greater penalty, thereby generating a larger error gradient during backpropagation. This intervention prevents the model from converging toward a trivial baseline solution, where the network simply defaults to predicting the most frequent class, ensuring the network tries to discover the underlying patterns of the inputs.

3.2 Performance Evaluation Metrics

Model evaluation was conducted on an independent, subject-wise test set comprising 20% of the dataset, ensuring a completely unbiased assessment. To evaluate the performance of the architecture, we relied on a combination of quantitative metrics. Accuracy was initially calculated to measure the overall proportion of correctly classified instances. Alongside it, the Area Under the Receiver Operating Characteristic curve (AUC-ROC) was utilized to evaluate the model’s general capability to discriminate between positive and negative emotional states. However, this metric ultimately proved less indicative of performance in this context.

Therefore, the most crucial measurement for our analysis was the Macro F1-Score, which harmonizes precision and recall. This is particularly useful in situations where there is a strong class imbalance, a condition present in our dataset prior to rebalancing. This metric helped us to demonstrate the robustness of the model in recognizing both classes equally, despite the initial statistical inhomogeneity. To derive these metrics and contextualize the results, confusion matrices were generated. This allowed for a granular analysis of misclassifications, detailing the model’s sensitivity in identifying true class

instances (Recall) alongside the correctness of its estimations (Precision).

To understand which physiological signals and features drive the classification of positive versus negative emotions, we employed SHAPley additive explanations (SHAP). This framework provides a robust quantitative measure of each feature’s impact on the model’s predictions. Based on this analysis, we evaluated the results of both phases, deriving critical insights into the physiological markers that most influence the network’s decision-making process.

4 Experiments

4.1 Experiments Phase 1

The first phase served to establish a robust unimodal baseline, providing a fundamental reference point prior to the integration of additional physiological modalities. The primary objective of this initial step was to evaluate the predictive limits of a neural network relying exclusively on ECG data.

In the following subsections, different training sessions of the model are described, progressing from the initial baseline to the final, which represents the optimal configuration achieved within our computational constraints. For each step, the most significant results obtained are reported, alongside the hyperparameter tuning and training adjustments implemented to improve iteratively the selected evaluation metrics.

4.1.1 Iteration 1 - Baseline and Class Imbalance Impact

For the initial training session, the objective was to observe the baseline behavior of the 1D-CNN without any structural interventions on the data distribution. The network was initialized with the following hyperparameter configuration:

- **Learning Rate (LR):** 0.001
- **Batch Size:** 32
- **Epochs:** 20
- **Dropout:** 0.5

The global evaluation metrics on the test set yielded the following results:

- **Accuracy:** 0.7811
- **F1-Score:** 0.5309
- **AUC-ROC:** 0.6980

Table 1: Classification report for Iteration 1 (Baseline).

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.80	0.11	0.19	4500
Positive (1)	0.78	0.99	0.87	14400
Macro Avg	0.79	0.55	0.53	18900
Weighted Avg	0.79	0.78	0.71	18900

Table 2: Confusion Matrix for Iteration 1.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	480 (TN)	4020 (FP)
Actual Positive (1)	118 (FN)	14282 (TP)

Despite an apparent 78.11% accuracy, the severe dataset imbalance (14,400 positive vs. 4,500 negative samples) causes

the network to develop a strict majority class bias. The model defaults to predicting the positive class, evidenced by a near-perfect 0.99 Recall for class 1 at the cost of 4,020 False Positives. Conversely, it correctly identifies only 11% of actual negative emotions (Recall 0.11). These metrics demonstrate that the current setup fails to extract meaningful physiological patterns. Ultimately, the network bypasses the true feature extraction, defaulting to automatically assign the most frequent label rather than learning the underlying biosignal dynamics.

4.1.2 Iteration 2 - Architectural Adjustments and Weighted Loss

To explicitly overcome the majority class bias observed in the baseline, Iteration 2 introduced targeted adjustments. First, Batch Normalization was integrated after each convolutional layer to standardize outputs, preventing uncontrolled numerical fluctuations and ensuring stable learning. Second, the Learning Rate was reduced to guarantee a safer convergence trajectory, preventing the network from overshooting the global minimum. Third, Dropout was lowered to 0.3, preserving a higher overall learning capacity to capture minute biosignal details. Crucially, a Class-Weighted BCE Loss was implemented to heavily penalize minority class misclassifications, forcing the model to actively extract features across both emotional states.

Table 3: Classification report for Iteration 2.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.39	0.47	0.43	4500
Positive (1)	0.82	0.77	0.80	14400
Macro Avg	0.61	0.62	0.61	18900

Iteration 2 shows a paradoxical 8% decrease in overall accuracy (70.07%), indicating that the network has stopped relying on majority class bias to actively analyze biosignal patterns. This is confirmed by the stable decline in Training Loss and, most importantly, by the Recall for Class 0, which increased from 0.11 to 0.47, quadrupling the model’s sensitivity to negative states.

This improvement introduced a necessary trade-off: Class 1 Recall decreased to 0.77 as the weighted loss forced the network to be more “cautious” before assigning positive labels, a prudence validated by the increased Precision for Class 1 (82%) and a higher Macro F1-Score (0.61). Although the AUC-ROC slightly decreased to 0.66, it reflects superior probability calibration, making the 0.5 decision threshold finally functional for both classes. This resolved the initial collapse toward a trivial solution, establishing a more reliable baseline.

4.1.3 Iteration 3 - Enhanced Regularization for High-Arousal Disambiguation

To overcome the intrinsic difficulty of distinguishing “High-Approach” emotions, which share similar high physiological arousal [3, 4], Iteration 3 introduced stricter regularization to force deeper temporal feature extraction. Dropout was increased to 0.4, and L2 Regularization (Weight Decay) was implemented to prevent single neurons from dominating the decision process, fostering distributed feature representation. To compensate for this harder learning process, the training duration was doubled to 40 Epochs.

The metrics confirm a successful optimization. Overall Accuracy increased to 72.63%, and the Macro F1-Score rose to 0.63, indicating a superior balance between the classes. The model successfully recovered approximately 500 positive instances previously misclassified, decreasing False Negatives from 3,290

Table 4: Classification report for Iteration 3.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.43	0.47	0.45	4500
Positive (1)	0.83	0.81	0.82	14400
Macro Avg	0.63	0.64	0.63	18900

to 2,796. Crucially, this was achieved without degrading minority class performance: Negative Class Precision increased to 43% while maintaining a solid Recall of 47%, leading to a significant improvement.

4.1.4 Iteration 4 - Dynamic Optimization and Loss-Based Checkpointing

Firstly, Iteration 4 introduced an LR Scheduler, which decreases the step size when the cost function is near the minimum, in order to avoid overshooting the optimal point and achieve the best possible approximation. Secondly, in this training session, we implemented Early Stopping and changed the checkpointing criterion: saving the model at the minimum validation loss rather than the maximum Accuracy.

Table 5: Classification report for Iteration 4.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.39	0.52	0.45	4500
Positive (1)	0.83	0.74	0.78	14400
Macro Avg	0.61	0.63	0.62	18900

Training stopped at epoch 30 with declining overall metrics (Accuracy: 68.98%, AUC-ROC: 0.67). While Class 0 Recall slightly improved (52%), the model lost nearly 1,000 True Positives. This degradation stems directly from the new saving criterion combined with the weighted loss: the network exploited a mathematical “shortcut”, over-predicting the negative class only to avoid heavy minority-class penalties and artificially minimize the global loss.

4.1.5 Iteration 5 - Random Undersampling and Definitive Baseline

To permanently resolve the class imbalance and prevent the evasive mathematical behavior observed in Iteration 4, Iteration 5 implemented a direct structural intervention: Random Undersampling. By randomly discarding approximately 66% of the majority class, the dataset was perfectly balanced (yielding ~20,000 samples per class). Furthermore, while retaining the LR Scheduler and Early Stopping, the checkpointing criterion was updated to save the model based on the maximum Macro F1-Score, forcing the network to optimize for predictive equilibrium rather than exploiting loss penalties.

Table 6: Classification report for Iteration 5.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.61	0.62	0.63	4500
Positive (1)	0.64	0.62	0.63	4800
Macro Avg	0.63	0.63	0.63	9300

The final evaluation reveals a nominal decrease in global Accuracy to 62.73%. However, in the context of an undersampled dataset, this is not a degradation but rather the complete elimination of the majority class bias. The model has achieved perfect predictive equilibrium: Precision, Recall, and F1-Scores for both classes are tightly clustered within the 0.61–0.64 range.

Table 7: Confusion Matrix for Iteration 5.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	2857 (TN)	1643 (FP)
Actual Positive (1)	1823 (FN)	2977 (TP)

Crucially, the model is now entirely devoid of statistical bias. Compared to the Iteration 1, where the network identified only 11% of the real negative emotions (480 TN), Class 0 Recall has nearly sixfolded to 63% (2,857 TN). The AUC-ROC consolidated at 0.6769, confirming robust class separability. Stripped of statistical artifacts, this honest and balanced configuration represents the true predictive limit of the unique ECG signal, officially establishing our definitive unimodal baseline.

4.1.6 SHAP Interpretability Analysis and Unimodal Limitations

To comprehensively evaluate the model’s decision-making process, a SHAP (SHapley Additive exPlanations) analysis [5] was conducted across four representative scenarios.

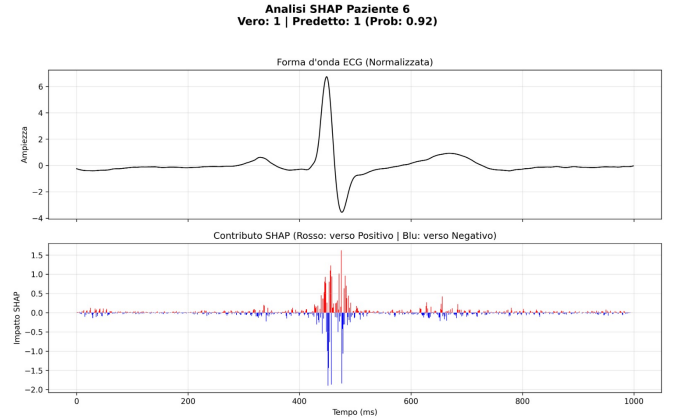


Figure 2: SHAP feature attributions for a highly confident correct prediction, clustered around the QRS complex.

In highly confident correct predictions (Figure 2), feature attributions cluster intensely around the QRS complex, confirming that the network correctly identifies and relies on core heart-beat morphology to drive its classification. However, the model struggles significantly with borderline instances (Figure 3 and Figure 4), where SHAP visualizations reveal highly contrasting attributions within the same signal. In these scenarios, competing features create extreme uncertainty: while in Figure 3 the model marginally achieves a correct prediction because negative contributions slightly outweigh positive ones, the identical ambiguity in Figure 4 results in a small misclassification (48% probability).

Crucially, in Figure 5, it is exposed the fundamental structural limitation of the unimodal ECG baseline through a high-confidence misclassification. Confronted with high-arousal emotions, the network is misled by the strong QRS amplitude, wrongly interpreting this intense physiological response as a positive state.

4.1.7 Conclusion and Transition to Phase 2

In conclusion, the optimization process yielded a robust, unbiased classifier that genuinely learns ECG morphological patterns, as transparently validated by SHAP analysis. Guided by the Macro F1-Score, the final balanced accuracy of 63% confirms that cardiac dynamics inherently contain valid emotional

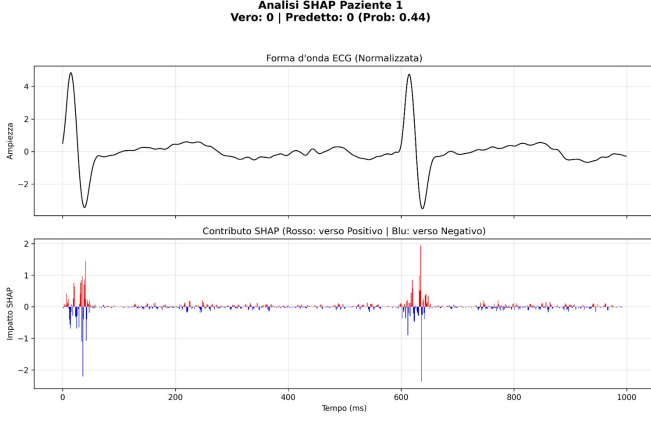


Figure 3: Borderline prediction with contrasting SHAP attributions (correct classification).

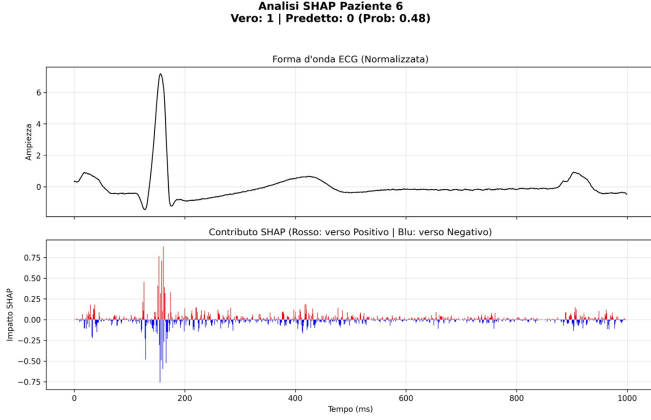


Figure 4: Borderline prediction with contrasting SHAP attributions (incorrect classification).

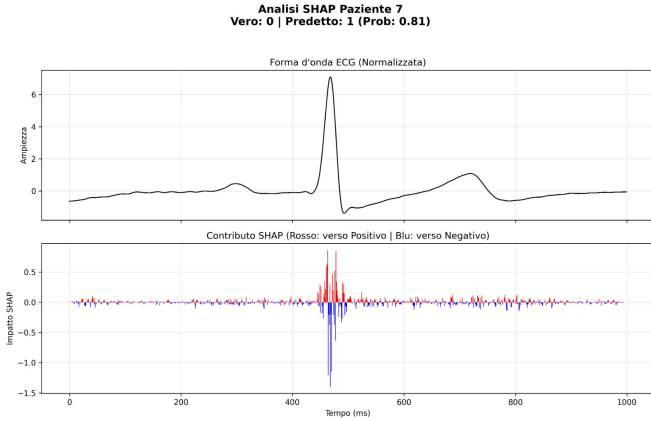


Figure 5: High-confidence misclassification demonstrating the limits of unimodal ECG in high-arousal scenarios.

information.

However, this percentage also exposes an intrinsic structural limit. As observed in the high-confidence misclassifications, this failure clearly proves that ECG alone cannot reliably distinguish between emotions with similar arousal levels, directly justifying the shift to multimodal architectures, using other biosignals to extract more information.

4.2 Experiments Phase 2

The second phase of the project marks the transition from unimodal analysis to a multimodal approach, integrating Electrocardiogram (ECG), Electrodermal Activity (EDA), and Affective self-report features. As demonstrated in Phase 1, while ECG provides foundational heartbeat morphology, it lacks the discriminative power to distinguish between emotions with similar arousal levels but opposite valence. Phase 2 explores two primary integration paradigms: Early Fusion and Late Fusion, aimed at capturing cross-modal correlations.

4.2.1 Early Fusion Architecture and Iterative Optimization

The Early Fusion strategy integrates multimodal data at the feature level. The vectors from the three sensors are concatenated into a single input layer, allowing the deep neural network to process them simultaneously. The architecture utilizes a deep linear structure with Dropout (0.4) for regularization and the Adam Optimizer. To refine convergence, a LR scheduler (step_size=10, gamma=0.5) was implemented to halve the learning rate every 10 epochs.

Version 1: Initial Unbalanced Baseline

The first iteration targeted a baseline behavior observation without structural interventions on data distribution. Although the model reached a peak validation accuracy of 78.38% at epoch 3, a subsequent divergence between training accuracy (83.61%) and validation accuracy (76.47%) indicated significant overfitting. The primary failure was the inability to detect negative emotions, with a critical Recall of 0.34 for Class 0.

Table 8: Metrics for Early Fusion - Version 1.

Metric	Version 1 Result
Epochs	20 (no early stopping)
Best validation accuracy	78.38% (epoch 3)
Global accuracy	59%
F1-Score (Macro Avg)	0.50
Precision (class 1)	0.77
Recall (class 0)	0.34

Table 9: Confusion Matrix for Early Fusion - Version 1.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	1535 (TN)	2965 (FP)
Actual Positive (1)	4709 (FN)	9691 (TP)

Version 2: Weighted Loss Management

To mitigate the majority class bias, Version 2 introduced a Weighted BCE Loss. This forced the model to penalize minority class misclassifications more heavily. While general sensitivity improved, the model maintained a statistical preference for positive instances.

Table 10: Metrics for Early Fusion - Version 2.

Metric	Version 2 Result
Accuracy	70.7%
Macro F1-Score	0.6391
AUC-ROC	0.7467
Recall (Class 0)	0.57

Table 11: Confusion Matrix for Early Fusion - Version 2.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	2582 (TN)	1918 (FP)
Actual Positive (1)	3620 (FN)	10780 (TP)

Version 3: Analysis of Extreme Dishomogeneity

This version tested the model against a severe imbalance (21,450 negatives vs. 65,400 positives) using Early Stopping at epoch 31. Despite a high AUC-ROC (0.7609), the model remained unstable, showing signs of a potential predictive collapse where the system tended to over-predict the negative class to minimize loss penalties.

Table 12: Metrics for Early Fusion - Version 3.

Metric	Version 3 Result
pos_weight factor	0.3280
Epochs	31 (with early stopping)
Global accuracy	68%
F1-Score (Macro Avg)	0.6357
AUC-ROC	0.7609

Version 4: Balanced Optimization (Definitive Baseline)

The final iteration implemented Physical Undersampling to reach a perfectly balanced dataset (21,450 samples per class). Combined with a dynamic learning rate reduction and Early Stopping (epoch 29), this version eliminated statistical bias.

Table 13: Global Metrics for Early Fusion - Version 4.

Metric	Version 4 Result
Epochs	29 (with early stopping)
Best validation accuracy	78.38% (epoch 3)
Global accuracy	66.4%
F1-Score (Macro Avg)	0.6630
AUC-ROC	0.7291

Table 14: Classification report for Early Fusion - Version 4.

Class	Precision	Recall	F1-Score
Negative (0)	0.63	0.73	0.68
Positive (1)	0.70	0.60	0.65

Table 15: Confusion Matrix for Early Fusion - Version 4.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	3270 (TN)	1230 (FP)
Actual Positive (1)	1899 (FN)	2901 (TP)

4.2.2 Late Fusion Architecture and Predictive Collapse Analysis

The Late Fusion approach relies on independent processing branches for Affect, ECG, and EDA signals, with the final decision governed by Majority Voting.

This architectural paradigm was tested to evaluate if modular independence could better isolate the specific contributions of each physiological modality, preventing the noise of one signal from polluting the feature extraction of the others.

Initial Training: Total Predictive Collapse

The first Late Fusion attempt (80 epochs, Dropout 0.4) resulted in a complete failure. The system achieved an accuracy of only 24%, reflecting exactly the statistical incidence of the minority class. The model classified 100% of instances as “Negative” (4500 TN, 14400 FN). Two critical factors caused this “lazy learning”:

- **Voting Rigidity:** Binary outputs (0 or 1) from branches eliminated the probabilistic granularity needed for mediation.
- **Aggressive Penalty:** Over-compensating for the imbalance led the network to a local minimum where over-predicting the negative class was the most “economical” way to minimize the loss.

Table 16: Classification report for Late Fusion - Initial Training.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.24	1.00	0.38	4500
Positive (1)	0.00	0.00	0.00	14400
Accuracy			0.24	18900
Macro Avg	0.12	0.50	0.19	18900
Weighted Avg	0.06	0.24	0.09	18900

Table 17: Confusion Matrix for Late Fusion - Initial Training.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	4500 (TN)	0 (FP)
Actual Positive (1)	14400 (FN)	0 (TP)

Optimized Late Fusion

A second session was conducted using Undersampling, increased Dropout (0.5), and three independent Learning Rate Schedulers (one per branch) to respect the specific complexity of each biosignal. This configuration successfully broke the local minimum observed in the previous iteration, allowing the modular branches to converge toward a balanced predictive state.

Table 18: Classification report for Optimized Late Fusion.

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.61	0.68	0.64	4500
Positive (1)	0.66	0.59	0.62	4800
Accuracy			0.63	9300
Macro Avg	0.64	0.63	0.63	9300
Weighted Avg	0.64	0.63	0.63	9300

Table 19: Confusion Matrix for Optimized Late Fusion.

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	3071 (TN)	1429 (FP)
Actual Positive (1)	1988 (FN)	2812 (TP)

4.2.3 Comparative Discussion

The experimental results confirm that Early Fusion is the superior strategy for this dataset, achieving a Macro F1-Score of 0.6630 compared to 0.63 for Late Fusion. The ability of Early Fusion to cross-reference data at a granular level (e.g., correlating localized ECG rhythm changes with EDA conductance spikes) provides a decisive informational advantage. Late Fusion, while more modular and resilient to individual sensor noise, suffers from the loss of inter-modal probabilistic information during the voting process.

The granular interaction allowed by Early Fusion enables the network to learn complex physiological signatures that a simple majority vote cannot capture. While Late Fusion offers greater architectural flexibility, the rigid nature of the decision-level integration proved to be a bottleneck for the nuanced task of emotional state classification. This concludes that for high-arousal biosignal disambiguation, a feature-level integration is essential to preserve the subtle cross-correlations between cardiac and skin conductance responses.

4.2.4 Multimodal SHAP Analysis

The SHAP analysis [5] for the multimodal Phase Two highlights a sophisticated hierarchical interaction between the integrated signals, where the ECG serves as the primary temporal anchor.

However, the EDA signal emerges as the critical “state-validator”; while its SHAP attributions are more temporally diffused, they provide the necessary stability to disambiguate high-arousal states that the ECG alone might misclassify. This dynamic is clearly visible in Figure 7, where the EDA contributes strong negative attributions that anchor the model toward the correct negative prediction. Interestingly, the AFFECT features maintain a lower-intensity, constant attribution profile, suggesting they provide a probabilistic baseline rather than decisive event-driven information.

The Early Fusion architecture successfully leverages this synergy: the model reaches peak predictive confidence only when the rapid cardiac dynamics of the ECG are corroborated by the tonic trends of the EDA. This multimodal interaction effectively mitigates the unimodal bias and achieves a more robust emotional equilibrium, allowing the network to balance conflicting signals even in highly uncertain borderline scenarios (as illustrated in Figure 6).

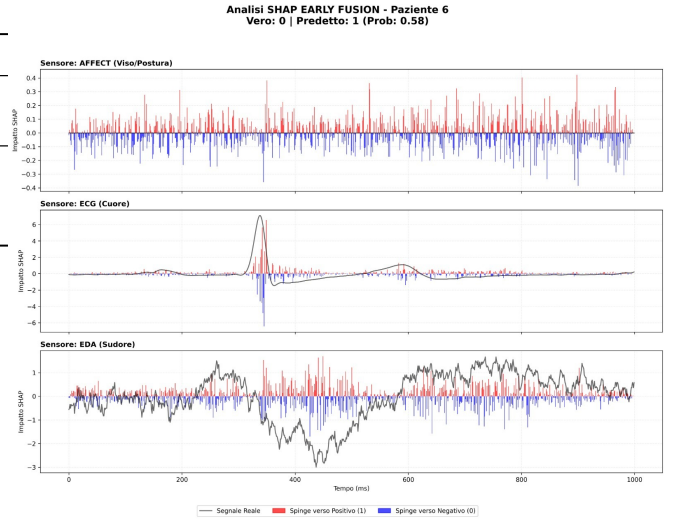


Figure 6: Multimodal SHAP interaction (Patient 6): A borderline scenario (Probability: 0.58) demonstrating the complex interplay where early fusion balances the strong, localized ECG spikes with the diffused attributions of the EDA and AFFECT signals.

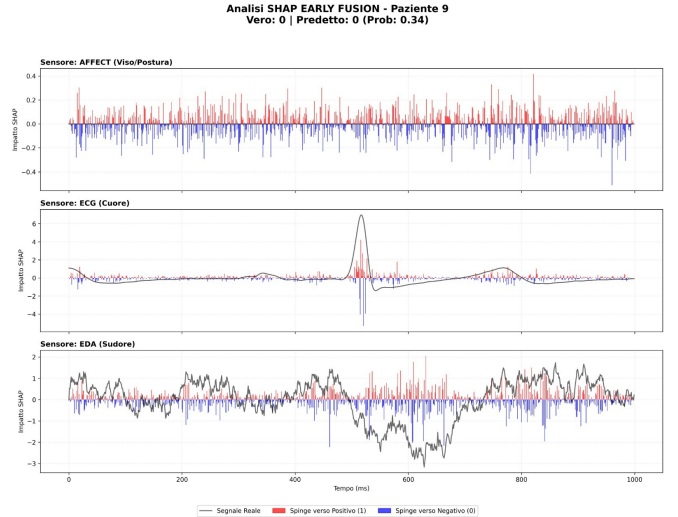


Figure 7: Multimodal SHAP interaction (Patient 9): The EDA signal effectively acts as a state-validator. Its strong negative attributions (blue bars) successfully guide the network toward the correct negative prediction (Probability: 0.34), compensating for ECG fluctuations.

5 Conclusions

This project successfully demonstrated that multimodal integration significantly outperforms unimodal ECG analysis in emotional classification, achieving a Macro F1-Score of 0.66. Beyond performance, the key learning outcome was the critical necessity of addressing dataset imbalance; we learned that undersampling is more effective than loss-weighting in preventing “lazy learning” and mathematical shortcuts. Challenges included resolving predictive collapses in Late Fusion, solved through independent learning rate scheduling.

Admittedly, expanding the dataset and extending the training phase, utilizing data augmentation to generate additional negative emotion instances, would yield further performance gains. However, despite the constraints of the current approach, the resulting classifier exhibits a good degree of robustness and reliability. Future extensions should aim to incorporate Recurrent Neural Networks (e.g., LSTMs or GRUs) to better capture long-term temporal dependencies, and explore real-time edge deployment in cyber-physical systems to critically assess latency-accuracy trade-offs.

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